

Table 3 — Comparison: Case Study A (Exploratory) vs. Case Study B (Canonical)
Case Study A identified the sensor column bug and API credit issue — both resolved for Case Study B

Metric	Case Study A (CMAPSS, June 9)	Case Study B (ETT-h1, 3 seeds, June 10)
Dataset	NASA CMAPSS FD001 (turbofan degradation)	ETDataset ETT-h1 (electricity transformer OT sensor)
Sensor column	setting_1 $\triangle$ (operational var., not sensor)	OT (oil temperature) $\checkmark$
n samples	20,631	17,420
MAE baseline	0.0018 $\triangle$ (near-zero — ADWIN insensitive)	1.6645 ± 0.1197 $\checkmark$
ADWIN detection	No detection ($ \Delta $ below Hoeffding bound) $\triangle$	$t=12,362$ — 27.4% into drift window $\checkmark$
Drift type	incremental (fallback heuristic, no ADWIN)	gradual (ADWIN-confirmed + decision tree) $\checkmark$
swap_trigger field	absent (pre-v2 run)	IMPROVEMENT ($\times 0.46$) $\checkmark$ [new audit field]
Copilot alert	FAIL (API credit error) $\times$	323–346 chars, consistent across all seeds $\checkmark$
Seeds / reproducibility	1 run (exploratory)	3 seeds — $\sigma=0$ on all key metrics $\checkmark$
Role in thesis	Debugging case — identified sensor col. bug	Canonical result — Case Study B in Section IV